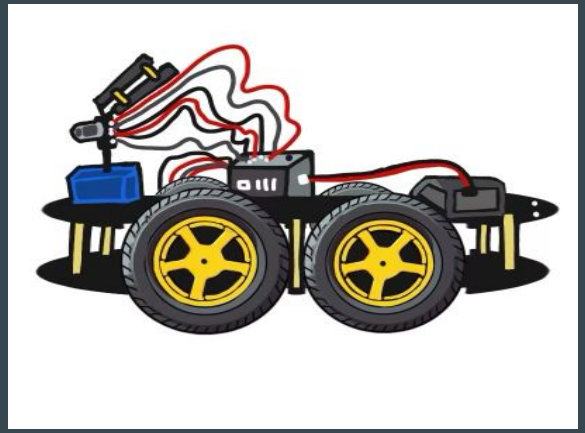
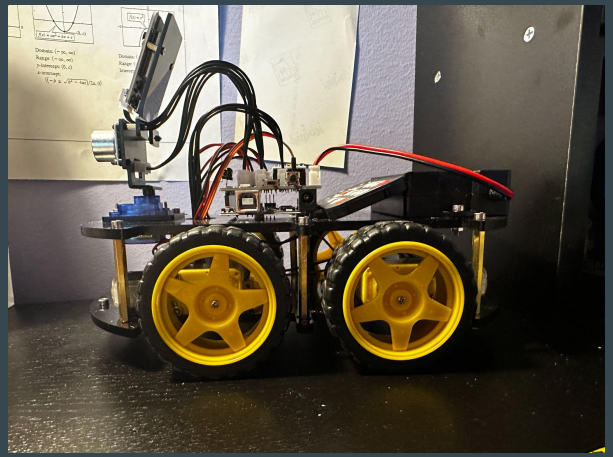
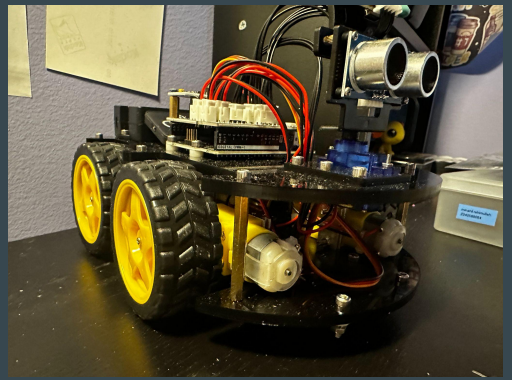


Measurement vehicle

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netID: mtino3, mRahi2 Group
Number: 69 Project
Name: Measurement vehicle





Measurement vehicle

Our project is a remote-controlled vehicle that can sense objects around it, allowing it to measure distance. We will use this feature to measure distance, allowing the user to pick between length, width, and distance modes. The first two modes measure the height and width of a given space, and distance mode allows the user to measure the distance traveled by the vehicle. All of this information will be displayed on the LCD on the controller.

Project Idea

The main project idea is a car that calculates the perimeter/area of a given space based on the distance traveled. This car is remote-controlled by the user, allowing for measuring different distances. This allows the user to measure length, width, perimeter, and distance traveled over time. The remote control will have three modes: length mode, width mode, and accessible mode. The length mode will allow the car to figure out the length of a specific space and return it on the LCD screen for the user to see. The width mode will do the same, and the accessible mode will constantly update the distance traveled by the car on the LCD as the user moves the car. The vehicle will come with a remote control so that the user does not have to pick it up and position it personally but can do so using the remote control.

Project Design – I/O Devices Used

- Joystick (input)
- Motor (output)
- Speed sensor (input)
- LCD display (output)
- Switch Buttons (input)
- nRF24L01+ transmitter/receiver (input/output)

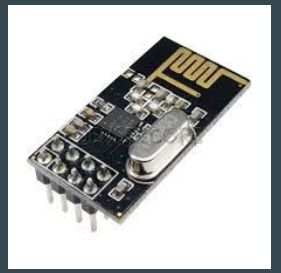


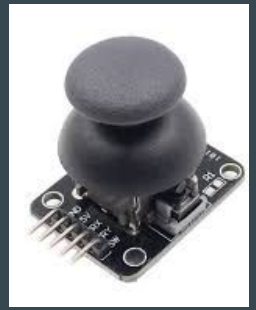






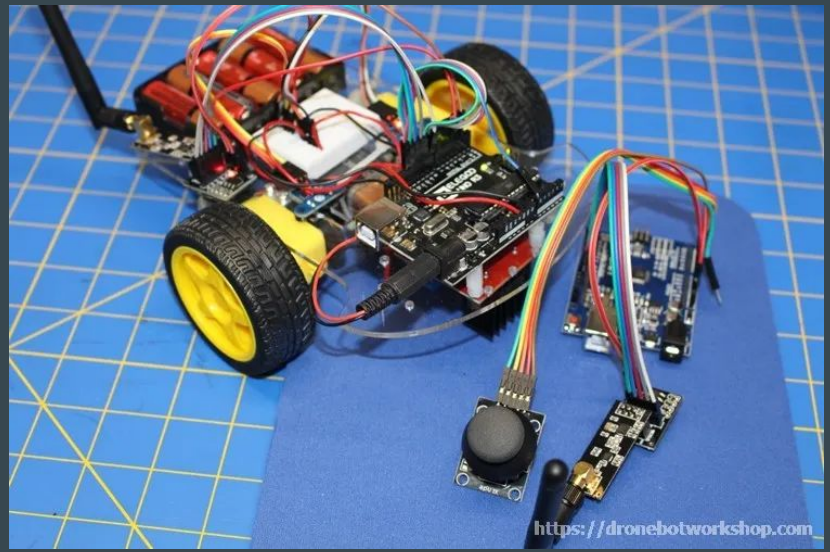






Project Design – Communication Used between Arduinos

- nRF24L01+ Module
  - Uses Radio waves to communicate
  - Better than infrared signals as they can pass through walls and objects
  - Using the adapter for this part extends the range from 100-200 meters to a few kilometers.
  - Prioritized range since there car is expected to be far from the remote control.



Project Design – Original work

We tried to make this idea unique by adding the measuring ability so that it has a practical feature and is interesting to implement. Our group understands many ways to create a four-wheeled vehicle, and these ways can be found online very easily. This is why we wanted to add this measuring ability so we could play around with the sensors, looking to see if we can measure distance based on how much the motors turn and see if we can make the Arduino calculate the distance between the sensor and the wall, joining the two together to find the distance between two walls.

What worked

- Using 2 wheels/motors - allows for a simpler circuit and allowed us to use the L298N motor driver to control the two wheels
- Using radio waves instead of infrared for communication - the signal goes through people and even walls and has a longer range, which is perfect for out build, allowing for the car to travel greater distances.
- Motors/L298N - figuring how to work the L298N was simple as each pin's responsibility was clear and not too complicated
- Speed sensors - did not give me trouble when wiring, and did what they said they were supposed to.
-

What doesn’t work

- Using four wheels - did a build with four wheels (on the front slide), but lead to no spots for the other components I wanted to add.
- Using infrared to communicate - my original design used infrared, however, this meant that you had to make sure the receiver was in direct line of sign of the remote, otherwise the signal would not go through, which lead to a laggy car
- Powersource - most trouble I had with this project was figuring how much or how little power I needed to put through this circuit, as I was piecing different circuits together and had to adjust accordingly, not to mention deciding if the arduino power source should be different or not.

Highlight process/team related roles

Tony - project coordinator

- Did most of project description + wrote out the details and list found on the milestones
- Figured out pseudo code for the project and helped with creating the file product

Minahil - project researcher

- Researched about the project, brought the parts together and assembled the circuit.
- Did wiring, experimenting, and troubleshooting of the project
- Looked at pseudo code and refined it with the help of partner to create the final product